1.62 Large Data Set: OCR MEI (London)

Worked Solutions

1 a)

32 plus the City of London

1 b)

9 areas: North East, North West, Yorkshire and The Humber, East Midlands, West Midlands, East of England, London, South East and South West

1 c)

The definitions of Inner and Outer London are those used by the Office of National Statistics and the Census.

2 a)

The percentage of people employed in the region with the age 16 to 64

2 b)

The amount of personal income per year.

2 c)

The percentage of all pupils in state schools at the end of key stage 4 who achieve five GCSEs at grade A* to C, including mathematics and English or better.

2 d)

The percentage of collected household waste which is recycled or composted.

3)

Houses which are sold for less than market value (such as right to buy houses).

4)

This data is estimated based on the Survey of Personal Incomes (SPI) which is an annual sample survey of HMRC records for individuals who could be liable to UK Income Tax.

5 a)

Simple random sampling

5 b)

Systematic sampling

5 c)

Convenience/opportunity sampling

5 d)

Stratified sampling (with simple random sampling).

6 a)

Mean is £686 867.80 sample standard deviation is £221 761.90 (population standard deviation is £210 381.81)

6 b)

Whilst there are a reasonable amount of areas selected from inner and outer London, the areas are predominantly from the top half of the data within Inner or Outer London (i.e. more expensive areas).

6 c)

686 867.80 $-2 \times$ 221 761.90 =£243 344 686 867.80 + $2 \times$ 221 761.90 =£1 130 391.60 (if using population: 686 868 $-2 \times$ 210 381.80 =£266 104.18 686 868 + $2 \times$ 210 381.80 =£1 107 631.42) There are no areas with median house price which are outliers.

6 d)

The median house price is much higher on average and more spread out in 2021 than in 2007.

7)

In both cases the median is less than the mean, which indicates a positive skew (more people with lower income). Kensington and Chelsea has a highly skewed income distribution with significant income inequality, characterised by a few extremely high earners that raise the average income far above the median. Lewisham has a more balanced income distribution with less income inequality and a median income closer to the mean.

8 a)

The top graph is Hackney.

8 b)

Hackney: $1.88 \times 12.5 + 26.3 = 49.8$ so £49 800 North West of England: $0.781 \times 12.5 + 23.3 = 33.0625$ so £33062.50

8 c)

Hackney: $26.3 \Rightarrow$ £26 300 is the mean income of someone in 2009/2010 1.88 is the rate at which mean income increases per year North West of England: $23.3 \Rightarrow$ £23 300 is the mean income of someone in 2009/2010 0.781 is the rate at which mean income increases per year

8 d)

According to the models, Hackney has a higher mean income in 2009/10 and has a higher rate of improvement in mean income than the North West of England.

9 a)

The City of London

9 b)

On average, outer London schools achieve the highest rates of percentage of all pupils getting 5 or more A*- C grades, with the rest of England performing slightly higher on average than Inner London. The rates of percentage of all pupils getting 5 or more A*- C grades are least consistent in Outer London, with similar consistency for Inner London and the Rest of England.

10 a)

The value of r indicates a moderate linear relationship between the recycling rate and the employment rate in 2011.

10 b)

$y = 0.979 \times 73.4 - 33.0 = 38.9\%$ (3 SF)

10 c)

It would lower the value of r The gradient of the regression line would be less steep and the y-intercept would translate upwards.

11 a)

$\mathrm{P}(X > 58) = 0.7580$ (4 DP)

11 b)

$\mathrm{P}(X < 55) = 0.0139$ (4 DP)

11 c)

$$\begin{aligned} &\mathrm{P}(59.4 - 2 \times 2 < X < 59.4 + 2 \times 2) \\ &= \mathrm{P}(55.4 < X < 63.4) \\ &= 0.9545 \text{ (4 DP)} \end{aligned}$$